Corrective Attention Paradigm

Peripheral-First Driving for Autonomous Vehicles

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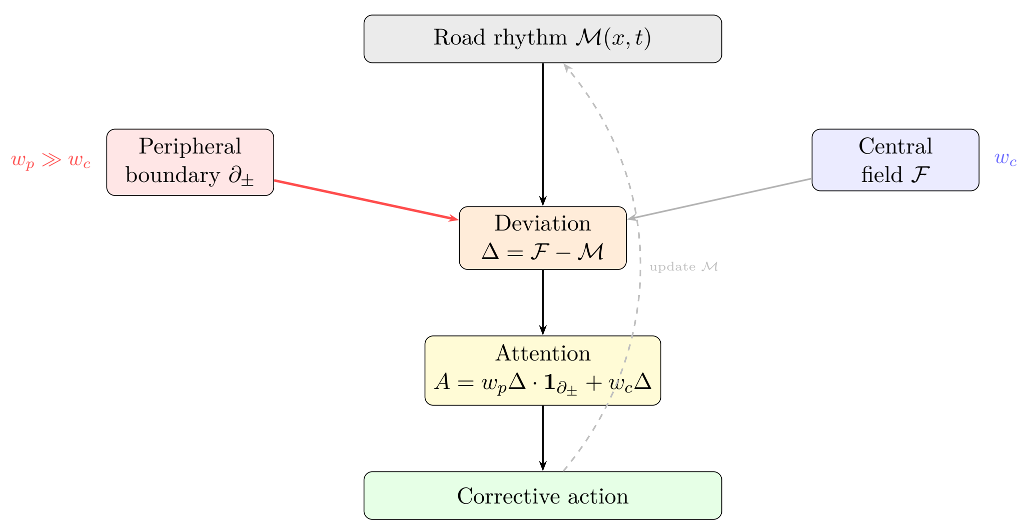
Abstract

Current autonomous vehicle systems weight central visual fields and lidar point clouds, with peripheral boundary detection treated as secondary. Human drivers invert this priority: peripheral polar boundaries — the edges between moving and stopped traffic, left/right contrast zones, the periphery of the visual field — are processed first, with central detail confirming. This inversion is not a limitation of human cognition. It is the correct architecture for danger detection. We propose the Corrective Attention Paradigm (CAP): an attentive conditioning framework that extends the social force model with temporal memory and peripheral-first weighting. The car attends to the rhythm of a specific road over time, corrects its behavior when the present deviates from learned pattern, and weights peripheral polar boundaries as its primary signal. CAP addresses the merge hesitation and edge-case failure modes characteristic of current center-weighted systems.

Architecture



1 The Social Force Model and Its Gap

Helbing and Molnár (1995) formalize traffic as a field of forces: each agent exerts attractive and repulsive forces on others, and motion is the response to the instantaneous force sum. The model captures following distance, lane keeping, and pedestrian flow with fidelity.

The gap is temporal. Social force models are reactive: they respond to current state. They do not attend to the pattern of states at a location over time. A merge on Highway 101 at 8:15am has a different rhythm than the same merge at 2pm. The instantaneous forces may be identical; the learned pattern is not. A human driver who uses that merge daily knows when to push and when to wait. The social force model does not.

2 Attentive Conditioning

Let $\mathcal{F}(x,t)$ be the social force field at location x and time t. Define the *attentive memory* at location x:

$$\mathcal{M}(x,t) = \int_{-\infty}^{t} e^{-\lambda(t-\tau)} \mathcal{F}(x,\tau)\, d\tau$$

where $\lambda > 0$ controls the decay rate. $\mathcal{M}$ is the exponentially-weighted history of forces at x — the road's learned rhythm.

The corrective signal is the deviation of present from learned:

$$\Delta(x,t) = \mathcal{F}(x,t) - \mathcal{M}(x,t)$$

A large Δ means the road is behaving unusually. The system attends to Δ, not to $\mathcal{F}$ alone. Correction is triggered by deviation from pattern, not by the raw force.

3 Peripheral-First Weighting

The human visual system detects motion and contrast at the periphery of the visual field before the fovea resolves detail. This is not a limitation — it is the correct priority. Danger arrives from the side. The front is where you are going. The sides are where it goes wrong.

Define the *polar boundary* $\partial_\pm$ as the spatial boundary between high and low velocity regions — the edge between moving and stopped traffic, the merge point, the lane boundary under contested entry. Formally:

$$\partial_\pm(x,t) = \{y : |\nabla v(y,t)| \text{ is locally maximal}\}$$

where v is velocity field.

CAP weights the attention field by proximity to $\partial_\pm$:

$$A(x,t) = w_c \cdot \Delta(x,t) + w_p \cdot \Delta(x,t) \cdot \mathbf{1}[x \in \partial_\pm^\epsilon]$$

where $w_p \gg w_c$ and $\partial_\pm^\epsilon$ is the ϵ-neighborhood of the polar boundary. Peripheral boundary events dominate the attention signal.

4 Failure Mode Analysis

Merge hesitation. Current systems hesitate at merges because the merge decision depends on gap acceptance — a judgment about a gap that is partially in the peripheral field, partially predicted from trajectory. Without peripheral-first weighting, the system waits for central confirmation. The gap closes. The hesitation compounds. Under CAP, the polar boundary at the merge point is the primary signal. The system attends to it directly and acts on the learned rhythm of gaps at this specific merge.

Edge case failure. The canonical autonomous vehicle failure: an object or behavior not in the training distribution appears at the periphery and is not detected in time. Under center-first weighting, peripheral anomalies are late. Under CAP, the polar boundary is monitored continuously and deviations from $\mathcal{M}$ trigger immediate attention shift. The edge case is still unknown, but its appearance at the boundary is noticed faster.

5 Organic Driving

The goal is not a car that follows rules. It is a car that participates in traffic — that drives *with* the system rather than against it. A car that knows it is in traffic, not above it.

Current autonomous systems drive defensively: they wait for certainty, which produces hesitation, which disrupts the flow they are trying to navigate safely. Organic driving is participatory: the car reads the rhythm, matches it, and corrects when the rhythm breaks.

CAP is the formal substrate of organic driving: attentive conditioning to local rhythm, peripheral-first detection of boundary events, correction by deviation from learned pattern.

6 Summary

	Current systems	CAP
Attention priority	center first	periphery first
Temporal model	reactive	attentive (learned rhythm)
Correction signal	raw force	deviation from pattern
Merge behavior	hesitates for certainty	reads gap rhythm
Edge case detection	late (peripheral lag)	fast (boundary monitoring)
Driving mode	defensive	participatory

References

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